MYOPIA: Understanding the Limits of Tree-Based Reasoning in Vision-Language Models

Supplementary Material

S1 Scope and Conventions

This supplement reports the complete results promised in the Reproducibility paragraph of the main paper. Section S2 gives every condition at both temperatures, including the tree-with-history ablation; Section S3 the paired-bootstrap intervals and corrected significance tests for the tree–zero-shot gap; Section S4 the ten-prompt zero-shot sweep; Section S5 the scope of repeated measurement; Section S6 the global-label probe; Section S7 the numbered-choice probe; and Sections S8–S9 the tree specifications, node counts, and audit coverage. Section S10 records the reproducibility details: the artifacts shipped with this supplement, the fixed image subsets, the decoding settings, and the limits on exact re-execution.

Three conventions apply throughout, and are stated here because they determine how every number below should be read.

Accuracy denominator. A prediction is correct when the integer ground-truth class equals the integer predicted class. The denominator is *every row of the condition*, so an unparseable answer and an image whose traversal was never logged both count as wrong. Accuracy therefore blends reasoning failure with answer-interface failure by construction; the two are separated only in the answer-interface analysis of the main paper and in Section S7 here.

GTSRB row deduplication. The GTSRB database contains legacy rows under `all_sequences_training/` that duplicate images stored under `datasets/GTSRB/Training/`. All tables in the main paper and in this supplement read through a view that drops the legacy copy whenever a matching modern row exists for the same model, temperature, ablation, and run. Ignoring this deduplication changes GTSRB denominators from 901 to as many as 915 and shifts accuracies by up to 0.3 points, so any recomputation that does not deduplicate will not reproduce these numbers exactly.

Coverage. A full pass is 901 images for GTSRB, 1000 for MNIST and CIFAR-10, and 850 for MSCOCO. The main paper prints a cell only when its condition covers at least 90% of a full pass. This supplement instead prints every cell that has data and annotates the incomplete ones with their n, so that what was and was not run to completion is visible rather than hidden.

S2 Complete Results at Both Temperatures

Table S1 reports all five prompting conditions at $T{=}0$ and $T{=}0.7$, extending Table 1 of the main paper with the $T{=}0$ block and the tree-with-history (Tree+H) ablation.

Two features of the table deserve comment. First, the $T{=}0$ block is complete only for GTSRB and CIFAR-10. On MNIST and MSCOCO the $T{=}0$ conditions were not run to completion, and the surviving cells cover between 2 and 860 images; they are printed with their n but should not be read as dataset-level estimates. This is the reason the main paper restricts its $T{=}0$ discussion to GTSRB and CIFAR-10.

Second, Tree+H is a full pass only on GTSRB and CIFAR-10. On MNIST and MSCOCO it covers 2–33 images, so the apparently strong values there (for example 75.00 for GPT-4o on MNIST, from 16 images) carry no weight. The history ablation should be read from the GTSRB and CIFAR-10 rows alone, where it reduces accuracy in five of six comparisons, the single exception being Qwen-VL on GTSRB.

Best and second-best marking considers only cells at full coverage, so a partial cell can never be marked as a winner.

S3 Significance of the Tree–Zero-Shot Gap

Table S1 shows the tree condition below zero-shot almost everywhere. Table S2 attaches an interval and a corrected p-value to that gap, so the claim rests on more than a point estimate.

The design is paired: for a given dataset, model, and temperature, the zero-shot and tree predictions for one image are stored on the same database row, so the two conditions see identical images. The bootstrap therefore resamples *images*, not predictions, and recomputes both accuracies on each resample; the reported interval is over the difference, which absorbs the correlation between the two conditions rather than treating them as independent samples.

Two limits are worth stating. The two-sided bootstrap p cannot fall below $2/10{,}000$, so eleven of the twelve comparisons report that floor and the corrected values for them are upper bounds rather than exact quantities. And the interval covers sampling variation over images at a fixed decoding temperature; it does not cover run-to-run decoding variation, which is the separate quantity measured in Section S5.

All twelve gaps are negative, and all twelve remain significant after Holm–Bonferroni correction over the family of

Table S1: Complete accuracy (%) for every condition at both temperatures. ZS: zero-shot; +D: class and image descriptions injected into the prompt; +H: conversation history carried across tree nodes; CoT: chain-of-thought. Accuracy counts unparseable answers and images with no logged traversal as wrong, so every cell shares the denominator of its condition. Superscripts give n for cells that did not complete a full dataset pass ($< 90\%$ coverage); an em dash marks conditions with no logged rows. CoT was run at T=0.7 only, on a n=100 probe per dataset–model pair with full class coverage. Bold and underlined values denote the best and second-best result within each dataset–model–temperature row, considering only cells at full coverage. Full passes are $n = 901$ (GTSRB), 1000 (MNIST, CIFAR-10), and 850 (MSCOCO).

Dataset	Model	T	ZS	ZS+D	CoT	Tree	Tree+H	Tree+D
GTSRB	GPT-4o	0	**70.92**	63.82	—	58.05	55.83	<u>66.81</u>
		0.7	**71.14**	63.60	63.00	57.71	56.71	<u>65.04</u>
	LLaMA	0	**23.86**	<u>18.42</u>	—	6.33	16.09	15.65
		0.7	**37.07**	23.53	12.00	<u>33.19</u>	12.43	14.87
	Qwen-VL	0	47.95	<u>49.72</u>	—	40.51	36.96	**51.83**
		0.7	47.06	<u>48.95</u>	36.00	24.64	36.96	**52.28**
MNIST	GPT-4o	0	94.87 ^{234}	88.02 ^{860}	—	35.47 ^{234}	46.67 ^{15}	—
		0.7	<u>91.50</u>	86.40	**94.00**	34.60	75.00 ^{16}	28.60
	LLaMA	0	33.33 ^{3}	—	—	33.33 ^{3}	0.00 ^{2}	—
		0.7	<u>59.30</u>	40.20	**71.00**	7.70	50.00 ^{2}	2.60
	Qwen-VL	0	95.17 ^{414}	—	—	14.25 ^{414}	—	—
		0.7	**94.40**	<u>89.20</u>	86.00	29.30	7.69 ^{13}	27.10
CIFAR-10	GPT-4o	0	<u>89.60</u>	**90.00**	—	75.40	73.10	82.70
		0.7	<u>87.90</u>	**88.90**	87.00	76.00	72.80	80.60
	LLaMA	0	<u>43.70</u>	**62.80**	—	17.10	12.30	4.80
		0.7	36.70	**54.00**	<u>38.00</u>	16.40	14.10	6.00
	Qwen-VL	0	**77.60**	<u>71.20</u>	—	56.60	33.00	66.50
		0.7	**77.60**	<u>70.90</u>	63.00	56.60	33.00	66.30
MSCOCO	GPT-4o	0	73.91 ^{92}	—	—	46.74 ^{92}	0.00 ^{7}	—
		0.7	<u>72.82</u>	**80.82**	71.00	46.71	33.33 ^{33}	44.00
	LLaMA	0	50.00 ^{2}	—	—	0.00 ^{2}	0.00 ^{4}	—
		0.7	**23.76**	<u>21.06</u>	18.00	3.65	33.33 ^{6}	0.35
	Qwen-VL	0	76.47 ^{221}	—	—	40.27 ^{221}	—	—
		0.7	<u>74.82</u>	**78.35**	59.00	39.65	—	44.47

twelve comparisons (largest adjusted $p = 0.0278$, GTSRB with LLaMA). The single smallest gap, -3.88 points for LLaMA on GTSRB, is also the only one whose interval approaches zero ($[-7.33, -0.44]$); every other interval excludes zero by a wide margin.

S4 Ten-Prompt Zero-Shot Sweep

The main paper notes that VLMs are sensitive to prompt phrasing and reports zero-shot results from one prompt. This section gives the full sweep behind that choice. Ten paraphrases of the zero-shot instruction were written per dataset; they differ in register and verbosity but request the same output, a single class ID. The prompt index is stored as `RunId`, and prompt 0 is the one reported in Table 1 of the main paper.

Table S3 and Figure S1 give per-prompt accuracy for the six dataset–model pairs where all ten prompts were run at full scale. Two observations follow, and both qualify the main paper's zero-shot numbers.

Prompt sensitivity is large, not small. Across-prompt standard deviations range from 2.81 points (CIFAR-10, GPT-4o) to 16.97 points (GTSRB, Qwen-VL), and the max–min spread reaches 41.5 points on GTSRB for Qwen-VL and 31.1 points for LLaMA. Prompt choice is therefore a first-order factor in the zero-shot number, comparable in magnitude to the differences between models. The spread is driven partly by catastrophic prompts: Qwen-VL falls to 5.55% on one GTSRB paraphrase and 50.1% on one CIFAR-10 paraphrase, patterns consistent with format non-compliance rather than a graded loss of recognition ability.

Prompt 0 is the strongest GTSRB prompt, not a typical one. On GTSRB, prompt 0 gives the highest accuracy of all sweep prompts for all three models: 71.14 against a mean of 59.67 for GPT-4o, 37.07 against 16.57 for LLaMA, and 47.06 against 35.84 for Qwen-VL. On CIFAR-10 it is close to the mean for GPT-4o (87.90 against 87.08) but above it for Qwen-VL (77.60 against 71.01) and below it for LLaMA (36.70 against 42.13). The GTSRB zero-shot column of Table 1 is thus a best-prompt figure rather than a typical-prompt figure.

This matters for the central Tree–zero-shot comparison, because tree traversal was run only under prompt 0 (Section S5) and the tree prompt does not vary with the prompt index. The

Table S2: Paired bootstrap over images for the tree-minus-zero-shot accuracy gap at T=0.7 (no descriptions, no history, run 0). Zero-shot and tree predictions share an image row, so images are resampled with replacement (10,000 resamples, seed 42) and both accuracies are recomputed on each resample. Δ is Tree$-$ZS in points; the interval is the 2.5–97.5 percentile range of the resampled Δ. p_{Holm} is the two-sided bootstrap p after Holm–Bonferroni correction over all twelve dataset–model comparisons. Entries marked $<$ are floored by bootstrap resolution and are upper bounds. Every comparison is negative and every corrected p clears $\alpha = 0.05$.

Dataset	Model	n	ZS	Tree	Δ	95% CI	p_{Holm}
GTSRB	GPT-4o	901	71.14	57.71	-13.43	$[-16.43, -10.54]$	<0.0024
GTSRB	LLaMA	901	37.07	33.19	-3.88	$[-7.33, -0.44]$	0.0278
GTSRB	Qwen-VL	901	47.06	24.64	-22.42	$[-25.64, -19.20]$	<0.0024
MNIST	GPT-4o	1000	91.50	34.60	-56.90	$[-60.00, -53.70]$	<0.0024
MNIST	LLaMA	1000	59.30	7.70	-51.60	$[-55.00, -48.10]$	<0.0024
MNIST	Qwen-VL	1000	94.40	29.30	-65.10	$[-68.10, -62.10]$	<0.0024
CIFAR-10	GPT-4o	1000	87.90	76.00	-11.90	$[-14.50, -9.40]$	<0.0024
CIFAR-10	LLaMA	1000	36.70	16.40	-20.30	$[-23.80, -16.80]$	<0.0024
CIFAR-10	Qwen-VL	1000	77.60	56.60	-21.00	$[-24.10, -17.90]$	<0.0024
MSCOCO	GPT-4o	850	72.82	46.71	-26.12	$[-29.88, -22.47]$	<0.0024
MSCOCO	LLaMA	850	23.76	3.65	-20.12	$[-23.06, -17.18]$	<0.0024
MSCOCO	Qwen-VL	850	74.82	39.65	-35.18	$[-38.82, -31.41]$	<0.0024

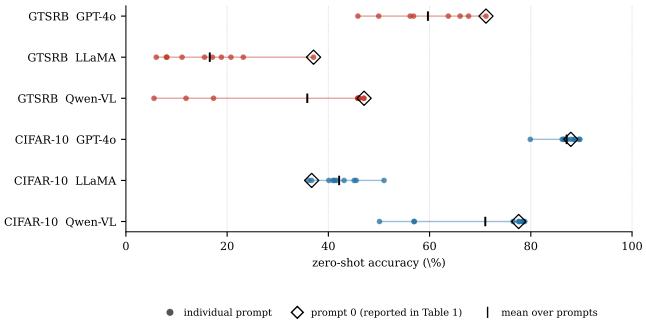


Figure S1: Zero-shot accuracy for each of the ten prompt paraphrases, on the dataset–model pairs swept at full scale. Prompt 0, reported in Table 1 of the main paper, is the best of the ten on GTSRB for all three models.

comparison therefore places a best-case zero-shot configuration against a single tree configuration on GTSRB. We report this explicitly rather than reselecting a prompt: the direction of the gap is unchanged, since even the weakest GTSRB zero-shot prompt for GPT-4o (45.84) is within 12 points of the tree result and the mean across prompts (59.67) still exceeds it, but the magnitude of the GTSRB gap should be read as an upper bound.

S5 Scope of Repeated Measurement

We state precisely what the repeated-measurement axis in our released databases does and does not cover, since this bounds the stability claims that can be made from them.

The `RunId` field selects the zero-shot prompt paraphrase. All ten paraphrases are distinct in every dataset. Consequently the variation quantified in Section S4 is *prompt* variation, and it applies to the zero-shot condition only.

Tree traversal was executed under `RunId` 0 alone; the tree prompt is built from the node's own question and answer set and does not depend on the prompt index. The database therefore contains one tree traversal per dataset–model–temperature–ablation cell, and no repeated-run replicate of a tree result exists at any scale. Reported tree accuracies are single-run measurements, and we make no claim about their run-to-run stability. Readers should treat the tree column of every table in the main paper and this supplement accordingly, and we regard replicated tree runs as the most important missing robustness check in this release.

Where a full-scale sweep is unavailable, Table S4 reports the small prompt checks that were run on MNIST and MSCOCO, restricted to the images all prompts share so the spread reflects prompt choice rather than differing image subsets. Three of the four checks are stable (SD $\leq$ 2.9 points), but MSCOCO with GPT-4o varies more (SD 4.79 points, range 65.0–75.0 on 20 images). These checks use 20 images each; their standard errors are wide and they do not substitute for a full-scale sweep.

S6 Global-Label Probe

Tree traversal shows the model only the current node's answer options, whereas zero-shot prompting shows it the entire label set. If that difference in label access were the primary cause of the Tree–zero-shot gap, restoring the global label list to each node prompt should recover much of it. This probe tests that, with GPT-4o, on class-stratified subsamples: each node prompt is issued twice, once in the standard local-list form and once with the dataset's full class list appended.

Table S5 reports both conditions on the images the two conditions share. The global list produces small and inconsistent changes in node accuracy: $+1.0$ point on GTSRB, $+0.1$ on MSCOCO, -1.4 on CIFAR-10 and -5.3 on MNIST. Image-

Table S3: Zero-shot accuracy (%) for each of the ten prompt paraphrases at T=0.7, on the datasets where every prompt was run at full scale. P0 is the prompt reported in Table 1 of the main paper. Empty cells are prompts whose run did not reach full coverage and are excluded from the summary statistics. Δ is $\max - \min$ across prompts.

Dataset	Model	P0	P1	P2	P3	P4	P5	P6	P7	P8	P9	Mean	SD	Δ
GTSRB	GPT-4o	71.14	63.71	—	66.04	45.84	56.16	67.70	49.94	—	56.83	59.67	8.94	25.31
	LLaMA	37.07	23.20	15.54	20.75	11.10	17.09	18.87	8.10	5.99	7.99	16.57	9.28	31.08
	Qwen-VL	47.06	46.95	46.06	45.84	5.55	45.84	46.06	11.88	17.31	45.84	35.84	16.97	41.51
CIFAR-10	GPT-4o	87.90	86.20	88.70	86.50	89.70	89.50	86.70	87.20	79.90	88.50	87.08	2.81	9.80
	LLaMA	36.70	51.00	43.10	36.10	41.20	41.60	40.90	45.10	40.10	45.50	42.13	4.38	14.90
	Qwen-VL	77.60	78.40	76.50	50.10	77.70	57.00	78.50	56.90	78.60	78.80	71.01	11.45	28.70

Table S4: Small-scale prompt checks on the datasets without a full-scale sweep. Each row compares the prompts that were run on the same matched image subset, so the spread is not confounded by which images each prompt saw.

Dataset	Model	n	Prompts	Mean	SD	Δ
MNIST	GPT-4o	20	4	93.75	2.50	5.00
	Qwen-VL	20	4	97.50	2.89	5.00
MSCOCO	GPT-4o	20	4	68.75	4.79	10.00
	Qwen-VL	20	4	68.75	2.50	5.00

Table S5: Global-label probe with GPT-4o, on the images both conditions share. Local is the standard node prompt; Global appends the dataset's full class list to every node prompt. All values are percentages.

		Node acc.		Unresolved		Image acc.	
Dataset	n_{img}	Local	Global	Local	Global	Local	Global
GTSRB	50	89.1	90.1	2.1	3.1	58.0	62.0
MNIST	20	71.1	65.8	0.0	0.0	35.0	35.0
CIFAR-10	20	88.9	87.5	1.6	7.8	65.0	60.0
MSCOCO	50	91.2	91.3	0.3	3.8	46.0	50.0

level accuracy moves by $+4$, $+4$, -5, and 0 points respectively. No dataset approaches its zero-shot accuracy from Table 1. The global list also slightly *raises* the unresolved rate on three of four datasets, consistent with the longer prompt making format compliance harder rather than easier.

We therefore find no support for label access as the primary cause of the gap. The probe uses 20–50 images per dataset, so it excludes only a large effect; differences of a few points are not resolvable at this scale.

S7 Numbered-Choice Probe

Because the strict extractor marks a response unresolved whenever it fails to match a predefined answer, part of the tree deficit could be an answer-interface artefact rather than a reasoning failure. This probe prospectively controls for that: on the same images, each node is asked twice, once with the free-form prompt and strict regex extraction, and once with an explicit numbered-option format that requests only an option number.

Table S6 reports both conditions. The numbered format eliminates unresolved responses entirely (0.0% on all four datasets), while free-form is already close to fully parseable for GPT-4o (0.0–2.4% unresolved). Node accuracy moves by at most a few points and not consistently in one direction.

Table S6: Numbered-choice probe with GPT-4o (n=20 images per dataset). Free is the free-form prompt with strict regex extraction; Num. requests only an option number. Node accuracy is computed over each condition's own visited nodes, as in the main paper. All values are percentages.

		Node acc.		Unresolved		Image acc.	
Dataset	n_{img}	Free	Num.	Free	Num.	Free	Num.
GTSRB	20	89.0	90.7	1.4	0.0	60.0	65.0
MNIST	20	68.3	71.4	2.4	0.0	35.0	30.0
CIFAR-10	20	92.3	92.2	1.5	0.0	75.0	75.0
MSCOCO	20	88.2	86.7	0.0	0.0	35.0	30.0

A caveat on matching. Each traversal terminates at its first incorrect or unresolved decision, so the two conditions do not visit the same number of nodes even on the same image: a condition that fails earlier accumulates fewer, shallower, and therefore easier nodes. The node accuracies in Table S6 are computed over each condition's own node set, which is what the main paper reports. Table S7 instead restricts to nodes reached at the same depth on the same image in both conditions. Matching in this stricter sense increases the MNIST gain from $+3.1$ to $+8.4$ points and deepens the MSCOCO loss from -1.5 to -3.8 points, while GTSRB and CIFAR-10 are essentially unchanged.

A caveat on power. The probe uses 20 images per dataset, yielding 36–110 node decisions per condition. None of the four differences is distinguishable from zero (Fisher exact, all $p > 0.75$), and at the image level the direction reverses on two datasets. The correct reading is that this probe excludes a *large* interface effect for GPT-4o, not that it measures a small one. Notably, free-form unresolved rates here are far below GPT-4o's rate in the full run (9.9% on GTSRB), which is what supports the main paper's claim that GPT-4o's interface failures are concentrated in particular conditions rather than inherent to free-form prompting.

Table S7: Numbered-choice probe restricted to nodes reached at the same depth on the same image in both conditions. Matching changes the MNIST and MSCOCO deltas relative to Table S6.

Dataset	Paired nodes	Free	Num.	Δ
GTSRB	73	89.0	90.4	+1.4
MNIST	36	69.4	77.8	+8.3
CIFAR-10	63	92.1	92.1	+0.0
MSCOCO	105	90.5	86.7	-3.8

S8 Tree Specifications and Node Counts

Table S8 gives the structure of the four evaluated trees. Every internal node has at least two children, every leaf maps to exactly one class, and no question repeats along any root-to-leaf path.

The trees differ substantially in shape. CIFAR-10 and MNIST are shallow (maximum depth 3) with 8–9 internal nodes; MSCOCO is an 89-node binary tree of depth up to 10; and GTSRB is the most irregular, mixing 16 binary nodes with two ternary, two five-way, and two eight-way nodes, and reaching depth 14. The eight-way arrow-direction question on GTSRB is the node an independent LLM judge flagged for question design, and it is also the highest-entropy split in any of the trees (3.00 bits).

Five MSCOCO internal nodes were never reached by any primary-condition traversal. They are excluded from the audit because they have no measured accuracy, which is why the audit covers 123 rather than 128 nodes.

S9 Tree-Quality Audit Coverage

Table S9 shows how the 347 node–model observations of the audit distribute over datasets and models. A node–model pair enters the audit only when it has both an in-traversal accuracy and a grounding gap; 1 of the 348 candidate rows lack one of the two and are dropped.

Coverage is strongly uneven. MSCOCO contributes 237 of the 347 observations (68%) from 83 nodes, while MNIST contributes 22 from 7 nodes. Any pooled association reported in the audit is therefore weighted towards the MSCOCO tree, which is also the only deep binary tree in the set. This is the main reason the audit's conclusions are limited to the four evaluated trees and are not presented as general properties of hierarchical classifiers.

Per-model coverage is 123 nodes for GPT-4o, 121 for Qwen-VL Max, and 103 for LLaMA-3.2; LLaMA reaches fewer nodes because its traversals terminate earlier. Node accuracy in the audit is thus conditional on reach, and nodes deep in a tree are scored on the subset of traversals that arrived there correctly.

S10 Reproducibility Details

This section records what a reader needs in order to regenerate the numbers above, and what cannot be regenerated exactly.

Artifacts in this supplement. The accompanying archive contains the four evaluated decision trees in full (both the human-readable and the machine-readable form), the complete prompt set (zero-shot paraphrases, per-class descriptions, and the node-question templates), the generator–evaluator tree-construction pipeline, the evaluation harness, and the analysis scripts that emit every table in this document. It excludes API credentials, raw provider traces, and the datasets themselves.

Sampling. Each dataset is evaluated on a fixed image subset, listed by identifier in the archive: 901 GTSRB, 1000 MNIST, 1000 CIFAR-10, and 850 MSCOCO images. The MSCOCO subset is restricted to images carrying exactly one category label, so that a single ground-truth class is well defined. These identifier lists are the sampling record; the subsets are not redrawn at run time.

Decoding. All primary conditions use temperature 0 or 0.7 with provider defaults otherwise, one image per request, and no few-shot examples. Tree traversal issues one request per visited node; the tree-with-history ablation additionally carries prior node questions and answers forward in the conversation.

Determinism and its limits. The analysis layer is deterministic: the bootstrap fixes seed 42 with $10{,}000$ resamples, and every table is recomputed from the stored databases rather than from cached summaries. The generation layer is not. All three models are served through hosted APIs, so even $T{=}0$ is not bit-reproducible, and the providers may retire or silently update a checkpoint. The LLaMA endpoint used for the main runs was in fact retired during the study and replaced by a self-hosted deployment of the same official weights at fp16. Re-running the pipeline should therefore reproduce the qualitative ordering and the sign of every gap in Table S2, but not the exact percentages.

Recomputation path. The stored databases are the ground truth for every table here. Given them, the analysis scripts in the archive regenerate this entire document, including the figure, without any API access or API key. Reproducing the databases themselves requires API credentials and paid calls.

Known pitfalls. Two details silently change results if missed. The GTSRB deduplication described in Section S1 must be applied or GTSRB denominators inflate from 901 to as many as 915. And accuracy counts unparseable answers and unlogged traversals as wrong; scoring only the parseable subset raises every tree number and is not comparable to the values reported here.

Table S8: Specifications of the four evaluated trees. Branching gives the multiset of child counts over internal nodes. Split entropy H_v is in bits, unnormalised. "Unreached" counts internal nodes that no primary-condition traversal ever visited.

Dataset	Int.	Leaves	Depth	Branching	$\bar{H}$	H range	Unreached
GTSRB	22	43	14	16×2, 2×3, 2×5, 2×8	1.02	0.35–3.00	0
MNIST	8	10	3	7×2, 1×3	1.01	0.92–1.30	0
CIFAR-10	9	10	3	9×2	0.96	0.81–1.00	0
MSCOCO	89	90	10	89×2	0.93	0.54–1.00	5
Total	**128**	**153**					**5**

Table S9: Tree-quality audit coverage. A node–model observation is retained when it has both an in-traversal accuracy and a grounding gap.

Dataset	Nodes	Node–model pairs
GTSRB	21	61
MNIST	7	22
CIFAR-10	8	27
MSCOCO	83	237
GPT-4o		123
LLaMA		103
Qwen-VL		121
Total	**123**	**347**